Design Patterns: Data Members & Relationships

Explicit UML Linkage Types, Fields, Method Signatures, and Constraints

1. Creational Patterns

Factory Method

UML Linkages: ConcreteCreator *Realizes (--->)* Creator. Creator has a *Dependency (- - ->)* on IProduct via creation return types. ConcreteCreator *Instantiates* ConcreteProduct.

Creator / ConcreteCreator Members
- defaultRegistry: Map<String, Class<?>> - logger: ILogger + factoryMethod(type: String): IProduct [Abstract] + anOperation(): void

Abstract Factory

UML Linkages: ConcreteFactory *Realizes (--->)* AbstractFactory. Client has a *Directed Association (---->)* to AbstractFactory. Concrete factories instantiate family product lines.

ConcreteFactory Members
- factoryCode: String - productConfig: Properties + createProductA(): AbstractProductA + createProductB(): AbstractProductB

Singleton Pattern

UML Linkages: Self-referential *Directed Association (---->)* pointing from Singleton class to itself via static reference field.

Singleton Members & Signatures
- uniqueInstance: Singleton [Static, Volatile] - instanceLock: ReentrantLock [Static] - birthTimestamp: Long - Singleton() [Private] + getInstance(): Singleton [Static] + performOperation(token: String): Boolean

2. Structural Patterns

Adapter Pattern

UML Linkages: Adapter *Realizes (--->)* Target interface. Adapter has an *Aggregation/Association (o---->)* holding a reference instance of Adaptee.

Adapter Members & Signatures
- adapteeRef: Adaptee - translationRules: Map<String, String> + request(data: Map<String, Object>): Response

Facade Pattern

Linkages: Facade has *Directed Associations (---->)* to multiple distinct Subsystem classes (SubsystemA, SubsystemB, SubsystemC).

Facade Members & Signatures
- subSystemAlpha: SubsystemA - subSystemBeta: SubsystemB - configOptions: ConfigModel + executeWorkflow(params: Bundle): Integer

Composite Pattern

UML Linkages: Composite and Leaf both *Realize (--->)* Component interface. Composite has a *Composition/Aggregation (<>----)* multi-valued association back to Component list.

Composite Members & Signatures
- childNodes: List<IComponent> - nodeIdentifier: String + add(component: IComponent): void + remove(component: IComponent): void + operation(): void

3. Behavioral Patterns

Command Pattern

UML Linkages: Invoker maintains an *Aggregation (o---->)* queue of Command interfaces. ConcreteCommand has a *Directed Association (---->)* to Receiver.

ConcreteCommand Members
- receiverTarget: Receiver - stateSnapshot: byte[] + execute(): void + undo(): void

Memento Pattern

UML Linkages: Originator has a *Dependency (- - ->)* creating Memento objects. Caretaker has a *Directed Association (---->)* storing Memento instances.

Memento Members
- savedStatePayload: InternalState - versionTag: Integer ~ getStateData(): InternalState [Package]

Strategy Pattern

UML Linkages: Context has a *Directed Association (---->)* referencing the Strategy interface. ConcreteStrategies *Realize (--->)* the Strategy interface.

Context & Strategy Members
- activeStrategy: IStrategy - executionTimeoutMs: Long + setStrategy(strategy: IStrategy): void + executeContext(data: List<Record>): ResultStatus